

CerviXAI

Full run guide — FastAPI backend + Flutter app, step by step

This guide covers everything needed to run CerviXAI from a fresh checkout: starting the backend, connecting the Flutter app to it (emulator, physical device over USB, or same Wi-Fi network), creating an account, and using both prediction paths in the app — a patient-record risk assessment and a cell-image screening with an AI-generated plain-language report.

Project folder	C:\Users\user\Desktop\Cervical_Cancer_XAI
Backend	FastAPI, in src/api.py — run from the project folder
Flutter app	frontend\flutter_app
Accounts / history	Stored on the backend (SQLite file at data\cervixai.db)

1. Prerequisites (one-time check)

Python	3.10 or newer, with pip
Flutter SDK	Installed and on PATH (run <code>flutter doctor</code> to check)
A device to run on	An Android emulator, OR a physical Android/iOS phone with USB debugging, OR any phone/emulator on the same Wi-Fi network as your computer
adb	Only needed for the USB-cable option — ships with Android Studio's SDK platform-tools

2. Backend — one-time setup

Open Command Prompt:

```
cd /d C:\Users\user\Desktop\Cervical_Cancer_XAI
python -m venv .venv
.venv\Scripts\activate
python -m pip install -r requirements.txt
```

You only need to repeat this step if `requirements.txt` changes (for example, it now also needs `python-multipart`, `scikit-image` and `pillow` for the image screening feature — run `pip install -r requirements.txt` again if you set this project up before that was added).

3. Start the backend (every time)

```
cd /d C:\Users\user\Desktop\Cervical_Cancer_XAI
.venv\Scripts\activate
python -m uvicorn src.api:app --host 0.0.0.0 --port 8000
```

Expected output:

```
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000
```

Keep this Command Prompt window open while you use the app. Press Ctrl+C to stop it.

Test it from the same computer:

```
curl http://127.0.0.1:8000/
curl "http://127.0.0.1:8000/cancer-report?patient_id=0"
```

Or open `http://127.0.0.1:8000/docs` in a browser for the interactive API docs.

4. Connect the Flutter app to the backend

The app needs your computer's address so the phone/emulator can reach the backend. Pick the row that matches how you're running it:

Android emulator	Use 10.0.2.2 — the emulator's alias for your computer.
Physical phone, USB cable	Plug in the phone, then run the adb reverse command below and use 127.0.0.1.
Phone/emulator, same Wi-Fi	Find your computer's IP with ipconfig (look for "IPv4 Address", e.g. 192.168.1.23) and use that.

If using a USB cable, with the phone connected and unlocked:

```
set "ADB=C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe"
"%ADB%" devices
"%ADB%" reverse tcp:8000 tcp:8000
```

The phone must show as device (not "unauthorized" or "offline"). The reverse command should print 8000.

5. Run the app

One-time setup after pulling these changes:

```
cd /d C:\Users\user\Desktop\Cervical_Cancer_XAI\frontend\flutter_app
flutter pub get
```

Then run it, replacing the address with the one that matches section 4:

```
flutter run --dart-define=API_BASE_URL=http://127.0.0.1:8000
REM emulator example:
flutter run --dart-define=API_BASE_URL=http://10.0.2.2:8000
REM same-Wi-Fi example:
flutter run --dart-define=API_BASE_URL=http://192.168.1.23:8000
```

To build an installable APK instead of running from the debugger, use the same `--dart-define` flag with `flutter build apk`. The APK is written to `frontend\flutter_app\build\app\outputs\flutter-apk\app-release.apk`.

6. Using the app

Create your account

- Open the app — it goes to the Sign-in screen after the splash.
- Tap **Create a professional account**, fill in your name, username, email and a password (8+ characters), then submit.
- Accounts are now real backend accounts (not stored only on the phone), so pick a password you'll remember — you'll need it again if you sign out or reinstall the app.

Path A — Patient record assessment (text/data-based prediction)

- From the **Dashboard** tab, tap **Start assessment**.
- Enter a **Patient ID** from the dataset — any whole number from 0 to 857 — and tap **Generate assessment**.
- The result screen shows the overall cancer-risk prediction and percentage, the supporting Biopsy and HPV model results, the SHAP explanation (which patient factors drove the result), a clinical interpretation, and a recommended next step.
- If you're signed in, this assessment is automatically saved to your account's History.

Path B — Cell image screening (image upload prediction)

- Tap the **Screening** tab in the bottom navigation.
- Tap **Gallery** to pick an existing photo, or **Camera** to take one.
- To try it immediately without a real slide photo, use one of the sample cell images bundled at `dataset/sample_cytology_images\` in the project folder — copy one to your phone's photos first, or run the emulator and drag-and-drop it into the emulator window.
- Tap **Run screening**. The report gives a plain-language result (Normal / Abnormal), a confidence percentage, the closest matching cell pattern, and a short explanation of what visually stood out — written so it makes sense to a patient, not only a doctor.
- This screening is also saved to your History automatically when signed in.

Editing your profile, history, and signing out

- **Profile tab** → **Edit profile** lets you change your name and email; it saves to your account on the backend, so it's still there next time you sign in on any device.
- **History tab** lists every assessment and screening you've run, newest first. Pull down to refresh, swipe an entry left to delete it, or use the trash icon to clear all history.
- **Profile tab** → **Sign out** ends your session on this device; your account, profile, and history remain on the server for next time you sign in.

7. Daily quick checklist

1	Start the backend (section 3) and leave the window open.
2	If using USB, connect the phone and run <code>adb reverse tcp:8000 tcp:8000</code> .
3	Run the app (section 5) with the correct <code>API_BASE_URL</code> .
4	Sign in (or register the first time).
5	Run a patient assessment or an image screening, and confirm the backend window logs a 200 OK.

8. Troubleshooting

"Could not reach the server" in the app	Confirm the backend window is still running, and that the address in <code>API_BASE_URL</code> matches your setup in section 4 (127.0.0.1 needs <code>adb reverse</code> ; 10.0.2.2 is emulator-only; a LAN IP needs the same Wi-Fi and no firewall blocking port 8000).
Device shows "unauthorized" or "offline"	Unlock the phone, accept the "Allow USB debugging" prompt, and re-run <code>adb devices</code> . If it's still stuck: <code>adb kill-server</code> then <code>adb start-server</code> .
Login fails with "Incorrect username or password"	Accounts are backend accounts now — if this is a fresh backend (a new/empty <code>data\cervixai.db</code>), you need to register again; old app-only accounts from before this update no longer exist.
Image screening says the model is unavailable	Make sure <code>models\image_screening_model.pkl</code> , <code>image_screening_features.pkl</code> and <code>image_screening_classes.pkl</code> are present, and that <code>scikit-image</code> and <code>pillow</code> installed correctly in step 2.
<code>pip install</code> fails to parse <code>requirements.txt</code>	This is fixed as of this update (the file was accidentally saved as UTF-16); if you still see odd errors, re-save <code>requirements.txt</code> as plain UTF-8 text.
Camera doesn't open on Android	The app now requests the Camera permission the first time you tap Camera in Screening — allow it. If you previously denied it, enable Camera for the app in Android Settings → Apps → CerviXAI → Permissions.

CerviXAI is a clinical-support and educational tool. Predictions and screening results are not a confirmed medical diagnosis — always interpret them together with professional clinical judgement.